Explainable Machine Learning for Severity 4 Traffic Accident Detection

Vy Truong[1][0009-0003-9684-9323], Thu Le[2][0009-0008-3480-8311],
Minh Huynh[3][0009-0002-2058-1298], Binh Dang[4][0009-0009-2873-4763]

FPT University, Ho Chi Minh Campus, Vietnam
truongnoen1sg@gmail.com, thulvm@fpt.edu.vn, minh061206@gmail.com,
dangphuocbinh2005@gmail.com

Abstract. Traffic accident severity prediction is a critical challenge for intelligent transportation systems, where timely identification of high-severity incidents can accelerate emergency response and improve adaptive traffic management. Severity 4 in the US-Accidents dataset denotes the highest degree of traffic disruption rather than a measure of injury or fatality severity. This study frames Severity 4 detection as a binary classification problem and evaluates six machine learning models on the US-Accidents dataset [10]. From 409,421 balanced records comprising equal counts of Severity 4 and non-critical accidents, 73 features are engineered after excluding geographic coordinates to prevent location-memorisation effects. Logistic Regression, Random Forest, XGBoost, LightGBM, CatBoost, and a Multi-Layer Perceptron (MLP) are trained and compared under stratified hold-out evaluation. Random Forest achieves the highest weighted F1-score of 0.8010 and AUC-ROC of 0.8756, while all tree-based models attain Severity 4 recall of 0.87–0.88. LightGBM demonstrates stable generalisation with a cross-validated AUC of 0.8681 ± 0.0019 across five stratified folds. Post-hoc explainability via SHAP global attributions and LIME instance-level analysis on LightGBM identifies road segment length, temporal patterns, and meteorological conditions as the principal predictors of high-severity outcomes, offering actionable insights for traffic safety stakeholders.

Keywords: Traffic accident severity, Binary classification, Explainable AI, SHAP, LIME, Gradient boosting, US-Accidents dataset

1 Introduction

Road traffic accidents represent one of the leading causes of preventable death and injury worldwide. According to the World Health Organization, road crashes claim approximately 1.35 million lives annually, imposing enormous economic and social costs on societies [17]. Within the United States, the National Highway Traffic Safety Administration categorises accident outcomes by severity. The highest level, Severity 4, denotes major road blockages with extended disruption, a traffic-impact rating rather than an injury severity classification. Identifying such critical incidents accurately and promptly can substantially improve emergency dispatch prioritisation, dynamic traffic rerouting, and infrastructure investment decisions within intelligent transportation systems (ITS).

The US-Accidents dataset [10] enables large-scale accident analysis by combining traffic event streams with weather, period-of-day, and OpenStreetMap POI annotations. However, limitations recur in existing work: comparative evaluations often overlook critical-class sensitivity, and models using geographic coordinates may memorise accident hot-spots rather than learning the *conditions* that drive high-severity outcomes [11].

Addressing these gaps, the present study contributes a coordinate-free, comparative, and explainable empirical analysis of Severity 4 accident detection. A reproducible preprocessing pipeline explicitly excludes geographic coordinate features, compelling classifiers to learn from temporal, meteorological, and road-context signals rather than spatial memorisation artefacts. Six models spanning a broad complexity range are systematically evaluated on a balanced dataset of 409,421 records, with performance assessed across accuracy, weighted F1-score, critical-class recall, and AUC-ROC. Empirical results establish that Random Forest achieves the highest test-set performance, while all tree-based models maintain Severity 4 recall of 0.87–0.88, confirming strong sensitivity for the safety-critical class. Post-hoc SHAP and LIME analysis on LightGBM connects model predictions to domain-interpretable factors, providing mechanistic insights relevant to traffic safety practitioners and infrastructure planners.

The evaluation is structured around three methodological questions: whether temporal, meteorological, and road-context features alone are sufficient for reliable Severity 4 detection when geographic information is withheld; which model best balances critical-class recall, overall predictive performance, and interpretability; and which features most strongly determine high-severity predictions as revealed by post-hoc explainability analysis.

The rest of this paper is organised as follows. Section 2 reviews related work. Section 3 details the proposed methodology. Section 4 describes the experimental setup. Section 5 presents and discusses the results. Section 6 analyses model explainability. Section 7 concludes the paper.

2 Related Work

Large-scale, publicly available accident data are a prerequisite for reproducible severity-prediction research. The US-Accidents corpus [10,11] is the foundational resource for the present study. Moosavi et al. [10] constructed the dataset by aggregating real-time traffic event streams from MapQuest and Microsoft Bing, augmenting each record with weather observations from 1,977 airport-based stations, period-of-day labels, and 13 OSM-derived point-of-interest annotation types. Extended to approximately 7.7 million records, it remains the largest publicly available countrywide accident dataset for the contiguous United States;

prior to its release, most accessible corpora were restricted to a single city or state, substantially limiting generalisability [11].

Machine learning for accident severity classification. Early work on severity prediction employed decision trees, naïve Bayes classifiers, and logistic regression on small-scale datasets [1]. Subsequent studies adopted random forests and gradient boosting for their ability to capture non-linear interactions in structured data [4,18,2]. Dong et al. [7] demonstrated that gradient boosting paired with SHAP attributions yields both strong predictive performance and interpretable feature rankings. Several prior benchmark studies included geographic coordinates as input features, risking location-memorisation effects that the present work explicitly guards against by removing all spatial columns.

Explainable AI in safety-critical domains. The opacity of high-performing models is particularly problematic in safety-critical applications where practitioners require actionable explanations alongside accurate predictions [3,16]. SHAP [9] provides theoretically grounded feature attributions derived from cooperative game theory, while LIME [15] constructs locally faithful linear surrogates around individual predictions. Both methods have been applied to traffic severity analysis [12,7], with SHAP global rankings and LIME instance reports emerging as complementary tools for bridging the gap between model performance and practitioner trust.

3 Methodology

3.1 Problem Formulation

The US-Accidents dataset assigns each record one of four ordinal severity levels. We frame the task as binary classification of Severity 4 — the highest traffic-disruption level — versus all non-critical outcomes, enabling direct optimisation of critical-class recall, the metric that most clearly quantifies the cost of missing a high-severity incident.

Formally, let the labelled dataset be $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^{N}$, where $\mathbf{x}_i \in \mathbb{R}^d$ is the feature vector for accident i and $y_i \in \{0, 1\}$ is the binary label with $y_i = 1$ denoting Severity 4 and $y_i = 0$ denoting any non-critical accident (Severity 1, 2, or 3). The objective is to learn a classifier $f : \mathbb{R}^d \to \{0, 1\}$ that achieves high Severity 4 recall while maintaining competitive overall accuracy, reflecting the asymmetric cost of false negatives in safety-critical deployment.

3.2 Dataset Design and Binary Labelling

The full US-Accidents March 2023 release contains approximately 2.8 million records, of which only 204,710 carry Severity 4 labels, representing roughly 7.3% of all records. Training directly on this imbalanced corpus would bias classifiers towards the majority non-critical class, artificially inflating overall accuracy while suppressing Severity 4 recall [5]. To isolate genuine model discrimination ability from imbalance effects, balanced binary sampling is adopted rather than synthetic oversampling: all 204,710 Severity 4 records are retained and an equal random sample of 204,711 records is drawn uniformly from Severity 1, 2, and 3 combined, yielding $N = 409{,}421$ balanced records.

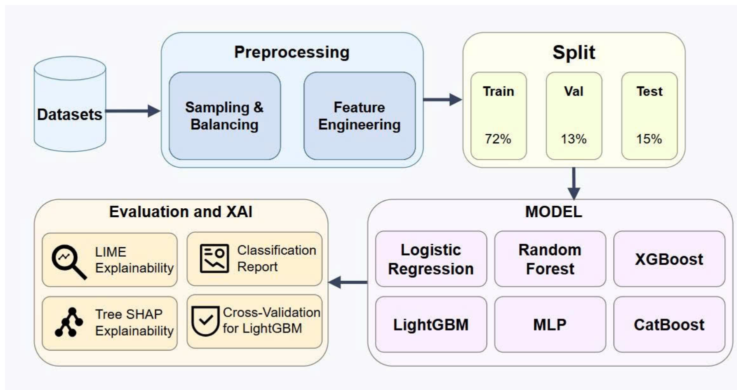


Fig. 1: Overview of the proposed methodology pipeline. Starting from the raw US-Accidents dataset, the workflow proceeds through sampling and balancing, coordinate-free feature engineering (73 features), stratified splitting, comparative model training, quantitative evaluation, and finally post-hoc XAI analysis via SHAP and LIME.

3.3 Feature Engineering Pipeline

The 47 raw US-Accidents columns are transformed into 73 model-ready features through four sequential steps. Each step is designed to maximise predictive signal while eliminating potential sources of data leakage and spatial overfitting.

Temporal feature extraction. Raw datetime strings carry implicit temporal context that direct encoding cannot expose. Five features are derived from the accident start timestamp: *hour* (0–23), *month* (1–12), *day_of_week* (0 = Monday), *is_weekend* (binary), and *is_rush* (binary; set to 1 during morning peak 07:00–09:00 or evening peak 16:00–19:00). These intervals exhibit markedly different Severity 4 distributions relative to non-critical accidents, justifying their inclusion as independent binary signals.

Geographic coordinate removal. Latitude and longitude columns are explicitly excluded to prevent location memorisation; models using them achieve inflated accuracy by remembering accident hot-spots rather than learning transferable predictors. *Distance(mi)*, which measures road-segment length rather than position, is retained.

Categorical encoding. The *Sunrise_Sunset* indicator is binarised into an *is_night* flag to provide a clean binary day/night signal. Three nominal features

— *Wind_Direction*, *Weather_Condition*, and *Timezone* — are one-hot encoded retaining the 20 most frequent categories per feature; the encoder is fitted exclusively on the training split to prevent information leakage into the validation and test sets.

Missing value imputation. Numerical columns with missing entries are imputed using training-set column medians, a robust choice for the skewed distributions typical of meteorological measurements. After all steps, each record is represented by $d = 73$ features, summarised in Table 1.

Table 1: Feature groups after engineering. OHE categories are fitted on the training split only to prevent leakage.

Feature Group	Count	Representative Features
Temporal	5	hour, month, day_of_week, is_weekend, is_rush
Meteorological	7	Temperature(F), Humidity(%), Visibility(mi), Wind_Speed(mph), Pressure(in), Wind_Chill(F), Precipitation(in)
Road context	2	Distance(mi), is_night
POI / Infrastructure	13	Junction, Traffic_Signal, Crossing, Stop, Roundabout, Railway, ...
OHE: Weather_Condition	20	Fair, Cloudy, Overcast, Rain, Snow, Fog, ...
OHE: Wind_Direction	20	Calm, N, S, E, W, NW, NE, SW, SE, ...
OHE: Timezone	6	US/Eastern, US/Central, US/Mountain, ...
Total	**73**	

3.4 Classification Models

Six models spanning a range of complexities are trained and evaluated: (i) Logistic Regression — ℓ_2-regularised binary logistic regression (SAGA solver) on standardised features, serving as the linear baseline; (ii) Random Forest — 300 trees with minimum leaf size 2; (iii) XGBoost [6] — histogram-based boosting (up to 2,000 trees, depth 6, learning rate 0.02, early stopping patience 50); (iv) LightGBM [8] — leaf-wise boosting (up to 2,000 trees, 63 leaves, learning rate 0.02, early stopping patience 50); (v) CatBoost [14] — ordered boosting (up to 500 iterations, depth 6, learning rate 0.045, early stopping patience 40); (vi) MLP [13] — a two-hidden-layer feed-forward network (256 and 128 neurons, ReLU activation, Adam optimiser, learning rate 0.001, batch size 1,024) on standardised features with internal early stopping (patience 15), serving as a neural network baseline.

3.5 Evaluation Metrics

Four metrics are reported per model: Accuracy, Weighted F1-score (harmonic mean of precision and recall weighted by class support), F1 (Severity 4) (per-class

F1 for the positive label, directly measuring critical-accident detection quality), and AUC-ROC (discrimination ability across all decision thresholds).

3.6 Explainability Methods

SHAP [9] decomposes each prediction into an additive sum of feature contributions: $f(\mathbf{x}) = \phi_0 + \sum_{j=1}^{d} \phi_j$, where ϕ_j is the Shapley value of feature j. We apply `TreeExplainer` on LightGBM to produce global bar plots, directional beeswarm plots, and feature dependence plots on a balanced sample of 600 test instances.

LIME [15] constructs a locally faithful linear surrogate g minimising $\xi(\mathbf{x}^*) = \arg\min_{g \in \mathcal{G}} \mathcal{L}(f, g, \pi_{\mathbf{x}^*}) + \Omega(g)$, where $\mathcal{L}$ is a locality-weighted fidelity loss and $\Omega(g)$ is a complexity penalty. Three representative instances are explained: a correctly predicted Severity 4 case, a correctly predicted non-critical case, and a false negative (Severity 4 misclassified as non-critical) — the error type with the greatest safety consequence.

4 Experimental Setup

Experiments use the US-Accidents March 2023 Kaggle release [10]. Following the preprocessing and feature engineering pipeline detailed in Section 3, the binary dataset comprises $N = 409{,}421$ balanced records and $d = 73$ features (see Table 1).

Data partitioning. The dataset is partitioned with seed 42 into stratified splits: 72% training (295,805 records), 13% validation (52,202), and 15% test (61,414), preserving 50% Severity 4 balance in every split. The validation set serves exclusively for early stopping in gradient boosting models; all metrics are reported on the held-out test set.

Hyperparameters. Hyperparameter values are set based on a combination of published defaults and preliminary validation-set probing. Logistic Regression uses ℓ_2 regularisation with `C=1.0` (SAGA solver, 1,000 iterations). Random Forest uses 300 trees with `min_samples_leaf=2` and `max_features='sqrt'`. XGBoost and LightGBM are both capped at 2,000 estimators with learning rate 0.02 and early stopping patience 50, which prevents overfitting without exhaustive grid search. CatBoost uses up to 500 iterations, depth 6, learning rate 0.045, and patience 40. The MLP uses two hidden layers (256, 128 neurons) with ReLU activation and Adam optimiser (learning rate 0.001, batch size 1,024), with scikit-learn's internal early stopping (patience 15) on a 10% internal validation fraction.

Cross-validation. LightGBM is additionally evaluated under 5-fold stratified cross-validation on the combined training-plus-validation pool of 347,007 records to assess stability across data partitions.

Implementation. All experiments are implemented in Python 3.10 using scikit-learn [13], XGBoost 1.7, LightGBM 4.x, CatBoost 1.2, SHAP, and LIME, all with `random_state=42` to ensure reproducibility.

5 Results and Discussion

Table 2 summarises test-set performance for all six models.

Table 2: Model performance on the held-out test set ($n = 61{,}414$, 50% Severity 4). Best value per column is bold.

Model	Acc.	Wtd F1	F1 (Sev4)	AUC
Logistic Regression	0.6635	0.6634	0.6597	0.7170
MLP	0.7687	0.7684	0.7760	0.8367
CatBoost	0.7883	0.7865	0.8059	0.8573
XGBoost	0.7950	0.7938	0.8093	0.8646
LightGBM	0.7964	0.7952	0.8108	0.8673
Random Forest	**0.8022**	**0.8010**	**0.8164**	**0.8756**

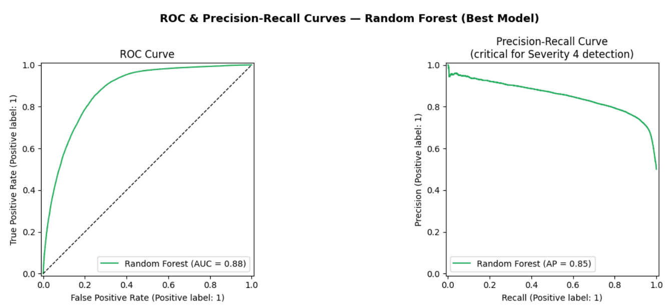


Fig. 2: ROC and Precision-Recall curves for Random Forest (AUC-ROC = 0.8756). The high-recall operating region of the PR curve confirms strong Severity 4 sensitivity at practical decision thresholds.

Random Forest as the best overall model. Random Forest achieves the highest scores across all four metrics (Table 2). As an ensemble of independently grown trees [4], it benefits from variance reduction without the sequential dependency of boosting, and its bagging mechanism proves well-suited to the heterogeneous, high-dimensional feature space after OHE expansion.

Gradient boosting cluster. LightGBM (F1 = 0.7952) and XGBoost (F1 = 0.7938) closely trail Random Forest, differing by less than 0.006 in weighted F1. Neither model fully converged within 2,000 iterations at learning rate 0.02, suggesting the classification surface in the coordinate-free feature space remains complex. CatBoost converges earlier (at iteration 451) and scores slightly lower (F1 = 0.7865), as its ordered boosting schema is more conservative on this balanced binary task.

Recall priority for Severity 4. All tree-based models achieve Severity 4 recall of 0.87–0.88 at the default threshold, with precision of 0.74–0.76. In safety-critical applications, false negatives (missed Severity 4 accidents) are more harmful than false positives; this recall-dominant operating point is therefore desirable, as further confirmed by the Precision-Recall curve in Fig. 2.

MLP as a neural network baseline. The MLP (F1 = 0.7684, AUC = 0.8367) outperforms Logistic Regression by 0.105 in weighted F1, confirming that non-linear transformations improve over linear models. However, it trails all four tree-based models by 0.02–0.03 in weighted F1 and converges in only 41 epochs, suggesting the feed-forward architecture quickly reaches its representational limit on this tabular feature space.

Linear baseline gap. Logistic Regression (F1 = 0.6634, AUC = 0.717) lags all non-linear models by at least 10 percentage points in weighted F1, confirming that Severity 4 prediction from non-geographic features requires modelling non-linear feature interactions.

Cross-validation stability. LightGBM achieves 5-fold stratified CV accuracy of 0.7960 ± 0.0017, weighted F1 of 0.7948 ± 0.0017, and AUC-ROC of 0.8681 ± 0.0019. The small standard deviations confirm stable generalisation across data partitions, validating that results are not artefacts of a specific split.

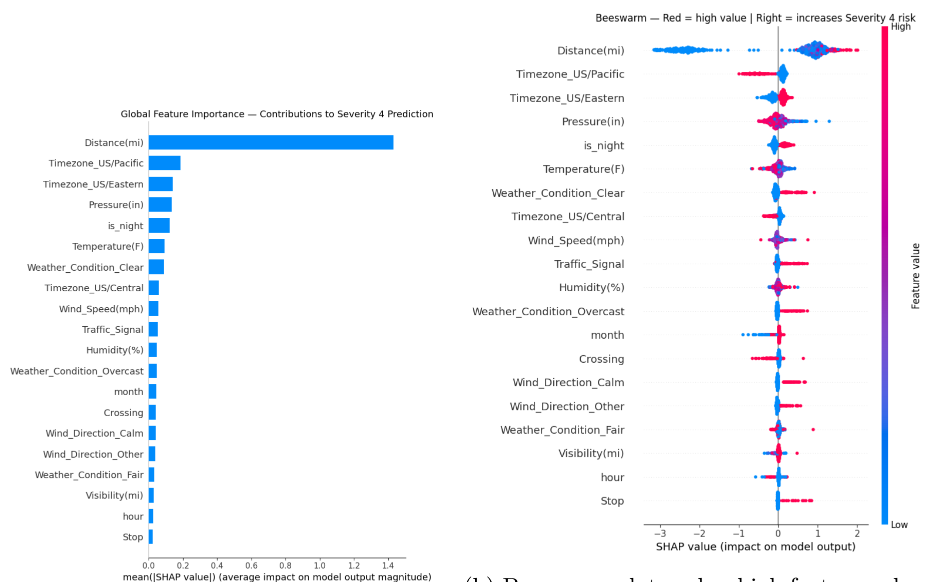
6 Explainability Analysis

Accurate predictions are necessary but insufficient for deployment in safety-critical domains. We apply SHAP and LIME to LightGBM, chosen for its native `TreeExplainer` compatibility and competitive performance (AUC = 0.8673, within 0.008 of the best model), to identify which features most strongly drive Severity 4 predictions and connect model behaviour to domain-meaningful traffic safety factors.

6.1 Global Feature Importance via SHAP

SHAP values are computed on a balanced sample of 600 test instances (300 Severity 4, 300 non-critical). Figure 3a ranks features by mean absolute SHAP value; Figure 3b adds directionality, where rightward displacement indicates positive contribution towards Severity 4.

Key findings. *Distance(mi)* emerges as the dominant feature — long-distance blockages strongly associate with Severity 4, consistent with more severe incidents disrupting larger road stretches. Temporal features (*hour*, *month*, *day_of_week*) contribute prominently, reflecting peak-hour and seasonal risk patterns. Meteorological variables (*Temperature(F)*, *Visibility(mi)*, *Humidity(%)*) corroborate prior evidence that adverse weather amplifies accident severity, with low *Visibility(mi)* consistently producing positive Shapley contributions. Among POI flags, *Junction* and *Traffic_Signal* rank among the top contributors, consistent with high intersection-accident rates [10].



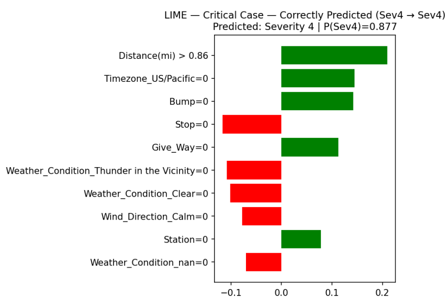
(a) Global feature importance (mean $|\phi_j|$). *Distance(mi)* dominates, followed by temporal and meteorological features.

(b) Beeswarm plot: red = high feature value; rightward shift = increases Severity 4 risk. High *Distance(mi)* and low *Visibility(mi)* both push towards Sev4.

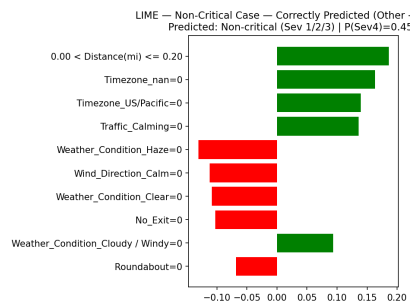
Fig. 3: SHAP global explanation for LightGBM. Road segment length, time-of-day, and visibility are the dominant drivers of Severity 4 predictions after geographic coordinates are removed.

6.2 Local Explanation via LIME

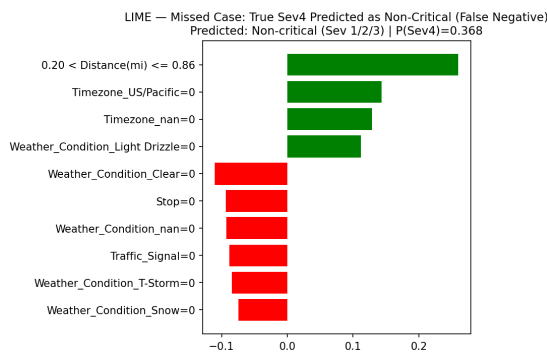
Figure 4 presents LIME explanations for three representative test instances, illustrating how feature contributions shift across prediction outcomes.



(a) Case 1: correctly predicted Severity 4.



(b) Case 2: correctly predicted non-critical.



(c) Case 3: false negative — true Severity 4 misclassified as non-critical.

Fig. 4: LIME explanations across three prediction cases. Positive weights (green) increase $P(\text{Sev4})$; negative weights (red) decrease it. Cases 1 and 2 show consistent feature alignment with correct outcomes, while Case 3 reveals that short *Distance(mi)*, daytime conditions, and no adverse weather collectively suppress the severity score below the decision threshold despite the true critical label.

Case 1 — Correctly predicted Severity 4 (Fig. 4a). Large *Distance(mi)*, low *Visibility(mi)*, and an evening hour collectively push $P(\text{Sev4}) > 0.80$, consistent with a highway incident under adverse weather at peak traffic.
Case 2 — Correctly predicted non-critical (Fig. 4b). Short distance, daytime conditions, and moderate visibility dominate the negative-weight features, yielding $P(\text{Sev4}) < 0.30$ — a profile unambiguously signalling a minor incident.
Case 3 — False negative (Fig. 4c). Short *Distance(mi)*, daytime conditions, and absent adverse-weather flags produce a profile resembling a minor accident, suppressing $P(\text{Sev4})$ below the threshold despite the true critical label — confirming

that some high-severity incidents share contextual signatures with non-critical ones when geographic location is withheld.

6.3 Practical Implications

The SHAP and LIME results, which agree on the top predictors (*Distance(mi)*, *hour*, *Visibility(mi)*), support three actionable conclusions: (*i*) large road-segment length events warrant high-priority dispatch alerts; (*ii*) real-time weather feeds can strengthen automated severity estimation by capturing adverse visibility and precipitation signals; and (*iii*) junction and traffic-signal proximity are targets for focused infrastructure safety improvement.

7 Conclusion

This paper presented an empirical, comparative, and explainable machine learning study for detecting critical (Severity 4) traffic accidents using the US-Accidents dataset. Six classifiers were trained on a balanced binary dataset of 409,421 records, with geographic coordinates explicitly removed to prevent location-memorisation bias. Random Forest achieved the highest performance (weighted $F1 = 0.8010$, $AUC\text{-}ROC = 0.8756$), with all tree-based models attaining Severity 4 recall of 0.87–0.88. LightGBM demonstrated stable cross-validated generalisation ($AUC = 0.8681 \pm 0.0019$) and was selected for post-hoc SHAP and LIME analysis. The explainability findings identify road segment length, temporal patterns, and meteorological conditions as the principal predictors of Severity 4 outcomes, providing actionable targets for traffic safety intervention.

Limitations and future work. *(i)* The binary design aggregates Severity 1, 2, and 3 into a single non-critical class; finer-grained severity ranking may yield additional insights. *(ii)* The dataset may under-represent rural accidents, introducing reporting bias towards urban corridors. *(iii)* The severity label reflects traffic flow impact rather than injury severity, limiting applicability to medical emergency contexts. *(iv) Distance(mi)* may itself be partially determined by incident severity; its dominant SHAP contribution should be interpreted as a predictive signal rather than an independent causal factor. Future directions include enriching features with road-type and speed-limit data, applying cost-sensitive learning or threshold calibration, and evaluating temporal train/test splits to better simulate real-world deployment.

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